

EDUCATION

The University of Hong Kong (HKU)

Ph.D., Electrical and Computer Engineering (ECE)

2024.09-2028.08

- **Advisors:** Prof. Evan Y. Peng, Prof. Edmund Yin Mun Lam
- **Research interests:** Computer Graphics (Simulation), Display

Tsinghua University (THU)

B.S. in Mathematics and Physics & B.Eng. in Measurement Control Technology and Instruments 2020.09-2024.07

- **GPA: 3.92/4.00, Ranking: 3/25.**
- Weiyang College at THU
- **Major Courses:** Fundamentals of Physics(1)&(2) (A+), Mathematical Physics Equations (A), Linear Algebra (A-), Electrodynamics (A-), Object-Oriented Programming (A-).

AWARDS & HONORS

- **National Second Prize (Silver Medal), 36th Chinese Physics Olympiad (CPhO)** 2019
- **Scholarship for Academic Excellence of Tsinghua University (1/25)** 2022
- HKU Presidential PhD Scholarship (HKU-PS) 2024
- Teaching Assistant Award, Department of ECE, HKU 2025

PUBLICATIONS

Boundary-aware Neural Model Reduction for PDEs 2026

*Li Liao**, Pengfei Shen*, Yifan Peng

SIGGRAPH 2026 (Technical Papers, to appear in July 2026)

Boundary-aware PDE eigenanalysis for reduced-order modeling, enabling unified spectral learning across varying geometries and boundary conditions.

Revisiting Hologram Generation, Compression, and Display with AI-Empowered Computational Techniques 2026

*Li Liao**, Wenbin Zhou*, Xiangyu Meng, Hyunmin Ban, and Yifan Peng

SID Symposium Digest of Technical Papers

S³ Imagery: Specular Shading from Scratch-Anisotropy 2025*Pengfei Shen, Feifan Qu, Li Liao, Ruizhen Hu, Yifan Peng*

SIGGRAPH Asia 2025 (Technical Papers)

Models scratch design as a constrained ODE under BRDF constraints to optimize anisotropic reflections for continuous view-dependent shading.

“Hands-on” Light Field Display via Engineering Scalable Optics, Algorithms, and Mounts 2025

Feifan Qu, Pengfei Shen, Li Liao, Yifan Peng

SIGGRAPH Asia 2025 (XR)

Empowering Head-up AR: Leveraging Holographic Display Engine, Geometry Optical Combiner, and Learned Calibration 2025

Wenbin Zhou, Xiangyu Meng, Li Liao, Yifan Peng

SIGGRAPH Asia 2025 (Emerging Technologies)

EXTRACURRICULAR

- **The 2019 voluntary activity to promote Tsinghua University to high schools.** Organizing and participating in information sessions, workshops, and presentations.

SKILLS

- **Programming** - C/C++, Python, MATLAB
- **Deep Learning** - PyTorch
- **Rendering** - Mitsuba, Blender
- **Optics** - Zemax